Ayushman Patro

Berhampur, Odisha

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Professional Summary

AI/ML Engineer experienced in Machine Learning, Generative AI, RAG, and Agentic AI, with hands-on work in model training, feature engineering, LLM applications, AI agents, and intelligent automation using Python, FastAPI, XGBoost, PostgreSQL, and cloud platforms.

Experience

Hutech Solutions | Gen AI Intern **6 Months**

- **AI-Powered E-commerce CMS:** Developed a multi-store content management platform using FastAPI, Next.js, Gemini LLM, Neon PostgreSQL, and Cloudinary, implementing AI-assisted content generation, RAG pipelines, embeddings, and dynamic editing of banners, product cards, tickers, and landing pages.
- **AI Voice Agent:** Built a conversational AI agent integrating LLM reasoning, speech I/O, prompt engineering, tool calling, and workflow orchestration to automate business operations and deliver context-aware responses.

Oasis Infobyte | Data Science Intern **3 Months**

- Developed and evaluated classification and regression models for spam detection, unemployment prediction, automobile-price estimation, and iris classification using EDA, preprocessing, feature engineering, model selection, and hyperparameter tuning.
- Improved model performance by 15% through cross-validation, systematic evaluation, feature optimization, and benchmarking of alternative ML algorithms.

Projects

AI-Powered Inventory Forecasting & Optimization System GitHub Live Demo **Oct. 2026**
Python, FastAPI, React, LightGBM, MILP, Docker

- Built time-series forecasting models and MILP-based inventory optimization, achieving 3.17% lower mean cost than the ROP baseline, with 132 automated tests.

Intelligent Patient Stratification & Drug Recommendation System GitHub Live Demo **Aug. 2026**
Python, XGBoost, Random Forest, SLM, Reinforcement Learning

- Developed patient-risk stratification and drug recommendation models using XGBoost, Random Forest, supervised learning, and ensemble methods.
- Built end-to-end ML pipelines covering EDA, feature engineering, model training, hyperparameter tuning, and cross-validation; integrated SLM reasoning and reinforcement-learning feedback loops.

Agentic Corporate Travel Automation Platform GitHub Live Demo **Jul. 2026**
Node.js, React, Travel APIs, Agentic AI

- Developed an AI-driven travel platform automating itinerary planning, travel recommendations, booking workflows, and multi-level approvals.
- Implemented natural-language processing, constraint-based planning, option ranking, tool orchestration, and plan-execute-validate agentic loops with human-in-the-loop approval.

Education

IIIT Bhubaneswar **Sep. 2022 – May 2026**
Bachelor of Technology in Computer Science *Bhubaneswar, Odisha*

Technical Skills

Programming: Python, JavaScript, TypeScript, Java, C, C++, SQL
Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Selection, Hyperparameter Tuning, Cross-Validation, Ensemble Learning, XGBoost, Random Forest, Reinforcement Learning, Model Evaluation
GenAI & LLM: LLMs, SLMs, Generative AI, RAG, Embeddings, Vector Search, Prompt Engineering, Context Engineering, LLM Application Development
Agentic AI: Agent Orchestration, Multi-Agent Systems, Tool Calling, Agentic Workflows, Agentic Loops, Human-in-the-Loop, MCP
AI Frameworks: LangChain, LangGraph, Spring AI, Hugging Face, OpenAI APIs, Gemini APIs
Backend & APIs: Node.js, FastAPI, Express.js, REST APIs, API Integration, Microservices, Backend Development
Frontend: React.js, Next.js, Flutter, MERN Stack
Databases & Cloud: PostgreSQL, MySQL, Vector Databases, Google Cloud, Render, Cloudinary
AI Developer Tools: Cursor, Claude, Google Antigravity, Kimi Code, Ollama, n8n, KIRO.dev
Developer Tools: Git, GitHub, Postman, VS Code, Google Colab, Figma

Certifications

Web Development: Full-Stack Web Development (Udemy)
Cybersecurity: Cybersecurity Fundamentals (Google)
Cloud: AWS Certified Cloud Practitioner
Generative AI: Generative AI Course Completion (LinkedIn)